

# **AAE 451 - Aircraft Conceptual Design**

## 3D-Printed Aircraft Design Course Notes

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**0.1 Lecture Notes**

### 0.1.1 Initial Vehicle Sizing by Constraint Analysis

#### Purpose of the Constraint Diagram

The correct sizing of an airplane depends on numerous important variables. A clearly stated mission plays a paramount role in this respect and allows the sizing to be accomplished using mathematical tools. Constraint analysis is used to assess the relative significance of selected aircraft performance parameters on the design with respect to a set of requirements.

Specifically, it allows one to understand which combinations of wing loading ( $W/S$ ) and power loading ( $P/W$ ) permit the aircraft to simultaneously meet dissimilar performance requirements, such as rate of climb, take-off distance, and others.

#### Overview of Constraint Analysis

Given mission requirements, we perform a constraint analysis using a constraint diagram. From this, we select design parameter values such as:

- Stall Speed ( $V_s$ )
- Cruise Speed ( $V_{cr}$ )
- Climb rate or climb angle
- Take-off run
- Maneuver loads

**Design Space:** The design space is that region within which all the constraints are met. For each quantitative requirement of the mission, we find the relationship between the wing loading and the power loading that exactly satisfies that requirement.

The general form of the Constraint Equation is:

$$\left(\frac{P}{W}\right)_{Sea-Level} = f\left(\frac{W}{S}\right) \quad (1)$$

For each requirement, we find the region where the vehicle violates that requirement and indicate the forbidden region with a hash mark. The feasible region (where we want to design our aircraft) is the intersection of all allowed regions. We typically prefer smaller engines (lower  $P/W$  for cost/weight) and smaller wings (higher  $W/S$ ), so the optimal point often lies at a corner of the feasible space.

#### 1st Constraint: Sizing to Stall Speed

To satisfy the stall speed requirement, the aircraft must be able to generate enough lift to balance its weight at stall velocity  $V_s$ .

$$L = W = \frac{1}{2}\rho V_s^2 S_{ref} C_{L_{max}} \quad (2)$$

Rearranging for Wing Loading ( $W/S$ ):

$$\frac{W}{S} = \frac{1}{2}\rho V_s^2 C_{L_{max}} \quad (3)$$

Where:

- $W$  = weight
- $S$  = wing area ( $S_{ref}$ )
- $\rho$  = air density
- $V_s$  = stall velocity
- $C_{L_{max}}$  = maximum lift coefficient

This is a vertical line on the constraint diagram. Any wing loading greater than this value will violate the stall speed requirement.

## 2nd Constraint: Sizing to Cruise Speed

Assuming the aircraft is in level flight during cruise, the power required ( $P_{req}$ ) must be less than or equal to the power available from the engine ( $P_{avail} \cdot \eta_p$ ).

$$P_{req} = D \cdot V_{cr} = \frac{1}{2} \rho V_{cr}^3 S C_D \quad (4)$$

Where:

- $\eta_p$  is the propeller efficiency
- $C_D$  is the drag coefficient

Cruise speeds for propeller-driven airplanes are typically calculated at 75 to 80 percent maximum power. Often, induced drag is small compared to profile drag, giving:

$$C_D = C_{D0} + 0.1C_{D0} = 1.1C_{D0} \quad (5)$$

If cruise is at sea-level, the constraint equation is:

$$\frac{P}{W} \geq \frac{1.1 \rho V_{cr}^3 C_{D0}}{2 \eta_p \left( \frac{W}{S} \right)} \quad (6)$$

This equation creates a curve on the constraint diagram where power loading must be greater than or equal to the required power loading for a given wing loading.

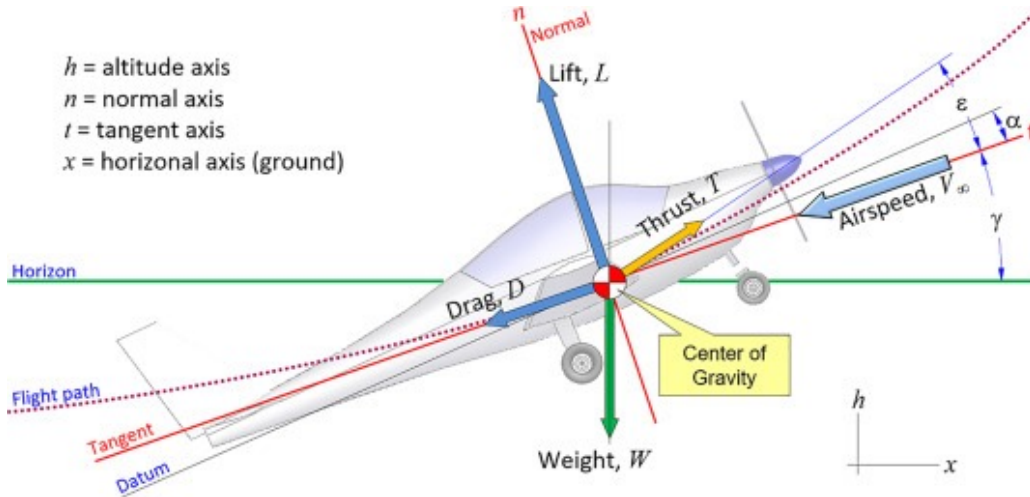


Figure 1: Constraint diagram showing the stall speed and cruise speed constraints with the feasible design region.

## 3rd Constraint: Sizing to Climb Requirement

For an aircraft in a climb:

- Thrust direction:  $T \cos(\alpha + \epsilon_T) - D = W \sin(\gamma)$
- Lift direction:  $L + T \sin(\alpha + \epsilon_T) = W \cos(\gamma)$

Assuming  $\alpha + \epsilon_T$  is small and  $T \sin(\alpha + \epsilon_T)/W$  is small, we have:

$$L \approx W \quad (7)$$

$$D = \frac{W}{L/D} \quad (8)$$

Climbing at  $L/D = 0.866(L/D)_{max}$ , the thrust required is:

$$T = \frac{W}{0.866(L/D)_{max}} + W \sin(\gamma) \quad (9)$$

Table 1: Variables for the Maneuver Constraint

| Variable                  | Description              |
|---------------------------|--------------------------|
| $q = \frac{1}{2}\rho V^2$ | Dynamic pressure         |
| $n$                       | Load factor              |
| $AR$                      | Aspect ratio             |
| $e$                       | Oswald efficiency factor |

Accounting for propeller efficiency  $\eta_p = \frac{T \cdot V_{climb}}{P}$ :

$$\frac{P}{W} \geq \frac{V_{climb}}{\eta_p} \left( \frac{1}{0.866(L/D)_{max}} + \sin(\gamma) \right) \quad (10)$$

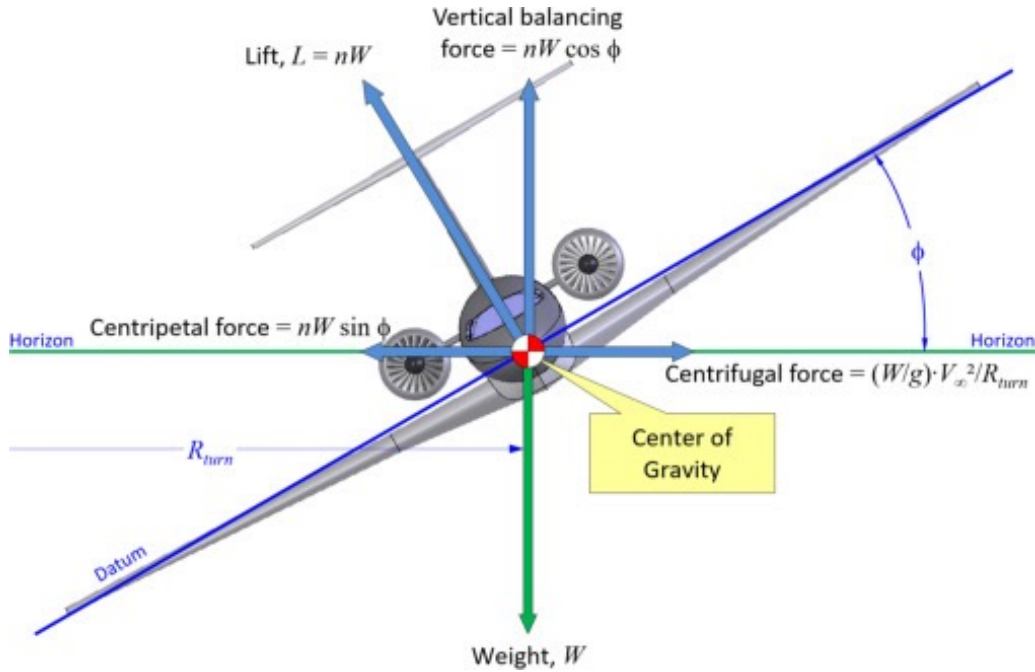


Figure 2: Constraint diagram including the climb constraint, illustrating how the design space narrows with each additional requirement.

#### 4th Constraint: Sizing to Maneuver Requirement

Consider an airplane in a level constant velocity turn. The load factor is  $n = 1/\cos(\phi)$ , where  $\phi$  is the bank angle.

The required thrust to maintain load factor  $n$  while banking at constant airspeed and altitude is:

$$T = qSC_{D_0} + \frac{n^2 W^2}{qS\pi eAR} \quad (11)$$

Dividing by weight and multiplying by velocity to get power:

$$\frac{P}{W} \geq \frac{V}{\eta_p} \left( \frac{qC_{D_0}}{W/S} + \frac{n^2(W/S)}{q\pi eAR} \right) \quad (12)$$

#### 5th Constraint: Take-Off

During the take-off run ( $S_{to}$ ), the aircraft accelerates from  $V = 0$  to  $V = V_{to}$  (typically  $1.1V_s$  to  $1.3V_s$ ). The general form for the take-off constraint will depend on runway friction ( $\mu$ ) and aerodynamic forces during the ground roll.

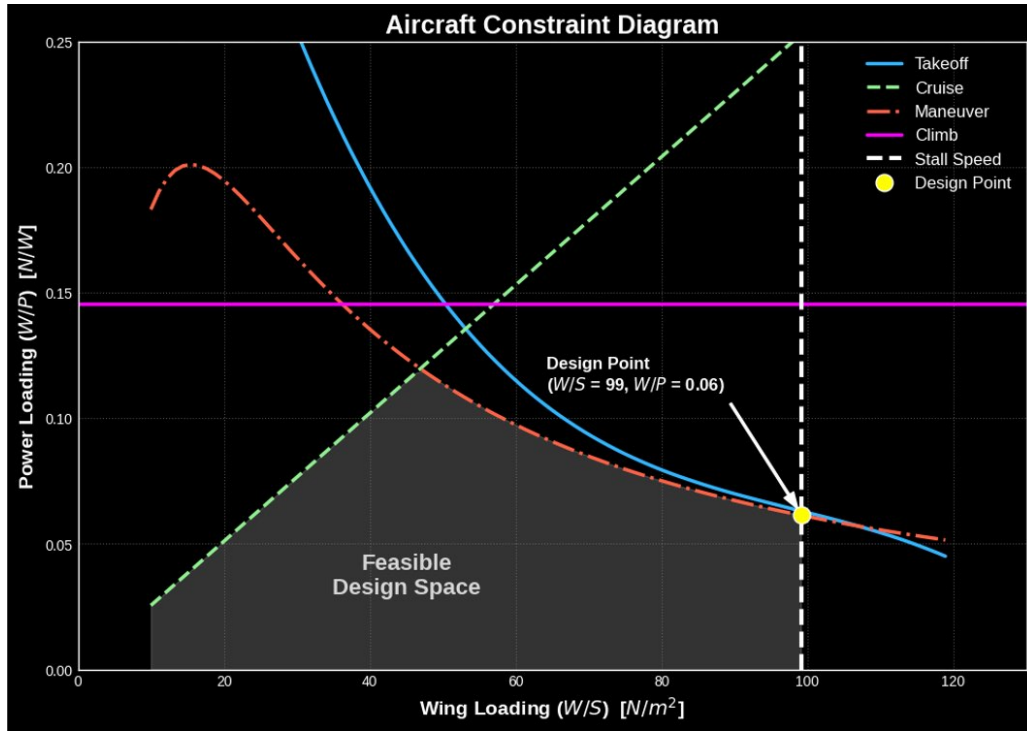


Figure 3: Complete constraint diagram with all four constraints (stall, cruise, climb, maneuver) showing the final feasible design space.

Table 2: Variables for the Take-Off Constraint

| Variable         | Description                                   |
|------------------|-----------------------------------------------|
| $S_{to}$         | Maximum take-off distance                     |
| $C_{Dg}, C_{Lg}$ | Drag and lift coefficients during ground roll |
| $\mu$            | Runway friction coefficient                   |
| $g$              | Gravitational acceleration                    |

Based on Sadraey's derivation (Chapter 4), the power loading constraint is:

$$\frac{W}{P} \leq \frac{\eta_P}{V_{to}} \left[ \frac{1 - \exp\left(\frac{0.6g\rho C_{Dg} S_{to}}{W/S}\right)}{\mu - \left(\mu + \frac{C_{Dg}}{C_{Lg}}\right) \exp\left(\frac{0.6g\rho C_{Dg} S_{to}}{W/S}\right)} \right] \quad (13)$$

### The Master Constraint Diagram

By plotting all the above constraints ( $P/W$  vs.  $W/S$ ), we identify the **Design Space**. The preferred design point is usually the one that minimizes the engine size (lowest  $P/W$ ) and minimizes wing size (highest  $W/S$ ) while staying within the feasible region.

**Note:** See the Jupyter Notebook `DBF_Constraint_Diagram.ipynb` for the Python implementation and visualization of these constraints.

### Reading & References

- **Gundlach:** Ch. 3, 9.4
- **Roskam (Vol. I):** Ch. 3
- **Gudmundsson:** Ch. 3
- **Sadraey:** Ch. 4

Table 3: Typical Mission Profile Phases

| Phase | Description                    |
|-------|--------------------------------|
| 1     | Warmup and takeoff             |
| 2     | Climb                          |
| 3     | Cruise (Level flight)          |
| 4     | Turning flight (if applicable) |
| 5     | Descent                        |
| 6     | Land and taxi                  |

Table 4: Weight Component Definitions

| Symbol | Description                                         |
|--------|-----------------------------------------------------|
| $W_e$  | Empty weight (structure, landing gear, motor, etc.) |
| $W_B$  | Battery weight                                      |
| $W_P$  | Payload weight                                      |

### 0.1.2 Initial Vehicle Weight Estimation

#### Our Approach

In this chapter, we focus on the initial weight estimation of electric-powered aircraft. The primary question is: **How much energy is needed to complete a design mission?**

To estimate this energy, our approach is as follows:

1. Consider how much energy is used in each mission phase.
2. Estimate the battery weight required to store that much energy.
3. Use historical data to determine the empty weight of the aircraft required to carry that battery plus any payload.

#### Weight Breakdown

The total aircraft weight ( $W$ ) is broken down into three main components:

$$W = W_e + W_B + W_P \quad (14)$$

#### Empty Weight Prediction

Empty weight ( $W_e$ ) consists of anything that cannot be removed from the aircraft without a tool. Early in the design process, when very little detail about the aircraft is known, we must use predictors or **Weight Estimating Relationships (WER)** based on historical data of similar classes of aircraft.

This historical data is usually represented as the empty weight fraction ( $W_e/W$ ) plotted against the total weight ( $W$ ), which typically yields a curve fit that we can use for estimation.

For a battery-powered, fixed-wing remotely controlled airplane, specific historical relations for  $W_e/W = f(W)$  can be established.

#### Power Flow

To understand energy consumption, we must first understand the flow of power from the battery to the aircraft.

Therefore, the total power required from the battery is related to the required thrust power by:



Table 5: Power Flow Chain

| Stage            | Power                    | Efficiency | Range   |
|------------------|--------------------------|------------|---------|
| Battery Output   | $P_B$                    | —          | —       |
| Motor Output     | $P_m = \eta_m \cdot P_B$ | $\eta_m$   | 0.7–0.9 |
| Propeller Output | $P_r = \eta_p \cdot P_m$ | $\eta_p$   | 0.5–0.9 |

$$P_r = \eta_p \cdot \eta_m \cdot P_B \quad \Rightarrow \quad P_B = \frac{P_r}{\eta_p \cdot \eta_m} \quad (15)$$

### Energy and Mission Phases

Energy ( $E$ ) is Power ( $P$ ) multiplied by time ( $t$ ):

$$E_{\text{battery}} = P_B \cdot t = \frac{P_r \cdot t}{\eta_p \eta_m} \quad (16)$$

We divide the mission into various flight phases and determine the energy required for each segment:

$$E_{\text{total}} = E_{to} + E_{wu} + E_{cl} + E_{lf} + E_{tf} \quad (17)$$

The total energy required by the battery ( $E_B$ ) determines the battery weight ( $W_B$ ) through the **energy density** of the battery ( $e_B$ , in J/kg; e.g., Lithium Polymer  $\approx 5.27 \times 10^5$  J/kg):

$$W_B = \frac{E_{\text{total}}}{e_B} \quad (18)$$

Since the individual energy components are functions of the total aircraft weight (which is yet to be determined), we normalize by the unknown aircraft weight  $W$ . The problem boils down to finding the **battery weight fractions** ( $W_{B,i}/W$ ) for each mission phase:

$$\frac{W_B}{W} = \frac{W_{B,to}}{W} + \frac{W_{B,wu}}{W} + \frac{W_{B,cl}}{W} + \frac{W_{B,lf}}{W} + \frac{W_{B,tf}}{W} \quad (19)$$

### Energy Consumption in Level Flight

In level flight, Lift equals Weight ( $L = W$ ) and Thrust equals Drag ( $T = D$ ). The thrust power required is:

$$P_r = T \cdot V = D \cdot V = \frac{W}{L/D} \cdot V \quad (20)$$

The energy required for a level flight time  $t_{lf}$  is  $E_{lf} = P_r \cdot t_{lf}$ . Factoring in the efficiencies and energy density, the battery weight fraction for level flight is:

$$\boxed{\frac{W_{B,lf}}{W} = \frac{V \cdot t_{lf}}{\eta_p \eta_m (L/D) e_B}} \quad (21)$$

### Energy Consumption in Turning Flight

In a turning flight with bank angle  $\phi$  and turning radius  $R$ , the load factor  $n = 1/\cos \phi$ . The lift increases, and consequently, drag increases by a factor of  $n$ .

$$\boxed{\frac{W_{B,lf}}{W} = \frac{V \cdot t_{lf} \cdot n}{\eta_p \eta_m (L/D) e_B}} \quad (22)$$

Table 6: Turning Flight Variables

| Variable                 | Description                |
|--------------------------|----------------------------|
| $R$                      | Turning radius (m)         |
| $\phi$                   | Bank angle (rad)           |
| $n = 1/\cos \phi$        | Load factor                |
| $g = 9.81 \text{ m/s}^2$ | Gravitational acceleration |

Table 7: Solution Procedure Steps

| Step | Action                                                                                        |
|------|-----------------------------------------------------------------------------------------------|
| 1    | For a given payload $W_P$ , calculate the total battery weight fraction $W_B/W$ from Eq. (19) |
| 2    | Compute battery weight: $W_B = (W_B/W) \times W$                                              |
| 3    | Rearrange the weight equation: $W - W_e = W_B + W_P$                                          |
| 4    | Plot $(W_B + W_P)$ and $(W - W_e)$ vs. total weight $W$ using the historical $W_e/W$ curve    |
| 5    | The intersection gives the <b>Initial Weight Estimate</b>                                     |

### Energy Consumption in Climbing Flight

In a climb to altitude  $h$  at climb angle  $\gamma$ , the sum of forces yields:

$$L = W \cos \gamma, \quad T = D + W \sin \gamma \quad (23)$$

Using  $D = L/(L/D) = W \cos \gamma/(L/D)$ , the power required is:

$$P_r = T \cdot V_{cl} = V_{cl} \cdot W \left( \frac{\cos \gamma}{L/D} + \sin \gamma \right) \quad (24)$$

For a time to climb  $t_{cl} = h/(V_{cl} \sin \gamma)$ , the battery weight fraction is:

$$\boxed{\frac{W_{B,cl}}{W} = \frac{V_{cl} \cdot t_{cl}}{\eta_p \eta_m e_B} \left( \frac{\cos \gamma}{L/D} + \sin \gamma \right)} \quad (25)$$

### Energy Consumption in Warmup and Take-off

During take-off for a time  $t_{to}$ , the battery output power is  $P_B = P_m/\eta_m$ . Normalizing by the airplane weight to get the inverse of the power loading ( $P_m/W$ ), the battery weight fraction for take-off is:

$$\boxed{\frac{W_{B,to}}{W} = \frac{t_{to}}{\eta_m (W/P) e_B}} \quad (26)$$

For warmup, a fraction  $N$  of the take-off energy is typically assumed:

$$\frac{W_{B,wu}}{W} = N \cdot \frac{W_{B,to}}{W} \quad (27)$$

**Solution Procedure Summary**

**Suggested Reading**

- **Gundlach:** Ch. 3
- **Raymer:** Ch. 3 and Ch. 20.11

Table 8: Wing Design Parameters to Determine

| Parameter       | Symbol                       | Description                             |
|-----------------|------------------------------|-----------------------------------------|
| Reference area  | $S_{ref}$                    | Wing planform area                      |
| Aspect ratio    | $AR = b^2/S$                 | Span-to-chord ratio                     |
| Taper ratio     | $\lambda = c_{tip}/c_{root}$ | Tip-to-root chord ratio                 |
| Sweep           | $\Lambda$                    | Leading or quarter-chord sweep angle    |
| Dihedral        | $\Gamma$                     | Upward angle of wing panels             |
| Twist (washout) | $\epsilon_t$                 | Spanwise twist angle                    |
| Airfoil(s)      | —                            | Cross-section profile                   |
| Incidence       | $i_w$                        | Wing setting angle relative to fuselage |
| Flap & aileron  | —                            | High-lift and lateral control devices   |
| MAC             | $\bar{c}$                    | Mean Aerodynamic Chord                  |

Table 9: High Wing vs. Low Wing Trade-offs

| High Wing                               | Low Wing                                |
|-----------------------------------------|-----------------------------------------|
| Increases $C_{L_{max}}$ (ground effect) | Better take-off performance             |
| Aircraft is laterally more stable       | Less frontal area                       |
| Produces slightly more lift             | Less induced drag                       |
| Usually lower stall speed               | More lateral control                    |
| Drag produces nose-down pitching moment | Less stable ( $C_{l_\beta} > 0$ )       |
| Less downwash on the tail               | More downwash effect on the tail        |
| Less lateral control                    | Less ground effect for take-off/landing |

### 0.1.3 Wing and Tail Design

#### Wing Design – Main Definitions

Lift comes with a price: **drag** and a **nose-down pitching moment**. Wing design decisions should therefore reflect the performance requirements, including stall speed, maximum speed, take-off run, range, and endurance.

#### Wing Arrangement

##### Number of Wings

- **Monoplane (single wing):** the practical and modern solution, enabled by advanced materials.
- **Biplane:** used when span is limited; shorter span delivers higher roll control, but at higher weight.

#### Wing Location

##### Airfoil Selection

The airfoil is the fundamental building block of the wing. Selection criteria must balance competing objectives.

**Wing Incidence** The wing incidence angle  $i_w$  should be set such that the fuselage generates **minimum drag during cruise**, i.e., the fuselage angle of attack is approximately  $0deg$  during the cruise condition.

Table 10: Airfoil Selection Criteria

| Objective                         | Design Driver                                 |
|-----------------------------------|-----------------------------------------------|
| Maximum lift ( $C_{l_{max}}$ )    | Take-off and landing performance, stall speed |
| Minimum drag ( $C_{d_{min}}$ )    | Range, endurance, max speed                   |
| Low pitching moment ( $C_{mac}$ ) | Reduced trim drag, smaller tail               |
| Gentle stall characteristics      | Safety and recoverability                     |

| Representative Aerodynamic Coefficients of Selected Airfoils (Low Re $\approx$ 200k) |             |       |        |                       |                 |                        |                  |                            |                 |  |  |
|--------------------------------------------------------------------------------------|-------------|-------|--------|-----------------------|-----------------|------------------------|------------------|----------------------------|-----------------|--|--|
|                                                                                      | Airfoil     | CLo   | CMo    | CL $\alpha$ (per rad) | CL,max (approx) | $\alpha_{stall}$ (deg) | Max L/D (approx) | $\alpha$ for Max L/D (deg) | CD,min (approx) |  |  |
| 0                                                                                    | Selig S1223 | 1.750 | -0.210 | 6.400                 | 2.200           | 15                     | 45               | 8.000                      | 0.019           |  |  |
| 1                                                                                    | Eppler E423 | 1.420 | -0.200 | 6.300                 | 1.850           | 14                     | 60               | 6.000                      | 0.012           |  |  |
| 2                                                                                    | Selig S1210 | 1.500 | -0.160 | 6.500                 | 2.000           | 16                     | 40               | 9.000                      | 0.018           |  |  |
| 3                                                                                    | Eppler 205  | 0.480 | -0.100 | 6.100                 | 1.450           | 13                     | 80               | 3.000                      | 0.009           |  |  |
| 4                                                                                    | NACA 2412   | 0.240 | -0.045 | 6.000                 | 1.550           | 15                     | 65               | 4.000                      | 0.008           |  |  |
| 5                                                                                    | SD7062      | 0.780 | -0.120 | 6.200                 | 1.500           | 14                     | 60               | 6.000                      | 0.010           |  |  |
| 6                                                                                    | Clark Y     | 0.360 | -0.050 | 5.700                 | 1.520           | 16                     | 70               | 4.500                      | 0.009           |  |  |
| 7                                                                                    | MH 32       | 0.220 | -0.057 | 6.200                 | 1.350           | 12                     | 90               | 2.500                      | 0.007           |  |  |

Figure 4: Representative aerodynamic coefficients of selected airfoils at low Reynolds number ( $Re \approx 200,000$ ), comparing lift-curve slope, maximum  $C_L$ , stall angle, maximum  $L/D$ , and minimum drag.

### Considerations about Aspect Ratio

The aspect ratio  $AR$  has a fundamental impact on the aerodynamic performance of the wing.

The **3D wing lift-curve slope** is related to the 2D airfoil lift-curve slope  $C_{l_\alpha}$  by:

$$C_{L_\alpha} = \frac{C_{l_\alpha}}{1 + \frac{C_{l_\alpha}}{\pi \cdot AR}} \quad (28)$$

The **induced drag factor** is:

$$K = \frac{1}{\pi \cdot e \cdot AR} \quad (29)$$

And the **maximum lift-to-drag ratio** is:

$$\left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{K \cdot C_{D_0}}} \quad (30)$$

Where  $C_{D_0}$  is the zero-lift drag coefficient, and  $e$  is the Oswald efficiency factor.

**Design rule:** The horizontal tail should have an  $AR$  lower than the wing to ensure the tail stalls after the wing (no twist required on the tail).

### Considerations about Taper Ratio

The taper ratio  $\lambda = c_{tip}/c_{root}$  influences the spanwise lift distribution, structural weight, and manufacturing complexity.

Table 11: Effects of Aspect Ratio

| Low AR                                 | High AR                           |
|----------------------------------------|-----------------------------------|
| Increases stall angle $\alpha_{stall}$ | Higher $L/D$                      |
| Reduces $C_{L_\alpha}$                 | Higher $C_{L_\alpha}$             |
| Lower structural weight                | Higher structural weight          |
| Better roll rate                       | Longer span, higher bending loads |

Table 12: Effects of Taper Ratio

| Effect                       | Description                                  |
|------------------------------|----------------------------------------------|
| $\lambda \approx 0.4 - -0.5$ | Approximates elliptical lift distribution    |
| $\lambda < 0.4$              | Reduces weight, but tip stall risk increases |
| $\lambda = 1.0$              | Rectangular wing — easy to manufacture       |
| Decreases $I_{xx}$           | Improves lateral control (roll rate)         |

Table 13: Tail Design Parameters

| Horizontal Tail           | Vertical Tail             |
|---------------------------|---------------------------|
| $S_t$ — planform area     | $S_v$ — planform area     |
| $b_t$ — span              | $b_v$ — height            |
| $l_t$ — tail arm          | $l_v$ — tail arm          |
| $\Lambda_t$ — sweep       | $\Lambda_v$ — sweep       |
| Airfoil                   | Airfoil                   |
| $AR_t$                    | $AR_v$                    |
| $\lambda_t$ — taper ratio | $\lambda_v$ — taper ratio |
| $i_t$ — incidence         | —                         |

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda} \quad (31)$$

### Considerations about Twist (Washout)

Wing twist (washout) reduces the angle of attack at the tip relative to the root. This serves two purposes:

1. **Prevents tip stall before root stall**, which is critical for maintaining aileron effectiveness and safe stall recovery.
2. **Helps approximate an elliptical spanwise lift distribution**, reducing induced drag.

A typical washout angle for general aviation is  $\epsilon_t \approx 2deg - -5deg$ .

### Tail Design

The horizontal and vertical tails provide pitch and yaw stability and control. The key parameters to determine are listed below.

### Pitching Moment and Longitudinal Stability

The total pitching moment about the center of gravity ( $M_{cg}$ ) is the sum of contributions from the wing, fuselage, propulsion, and tail:

$$C_{M_{cg}} = C_{M_{ac,w}} + C_{M_{fus}} + C_{M_{prop}} + C_{M_{tail}} \quad (32)$$

Expanding:

$$M_{cg} = M_{ac,w} + (x_{cg} - x_{ac})L_w + M_{ac,t} - l_t L_t + M_F + M_P \quad (33)$$

The **pitching moment coefficient** about the c.g. is:

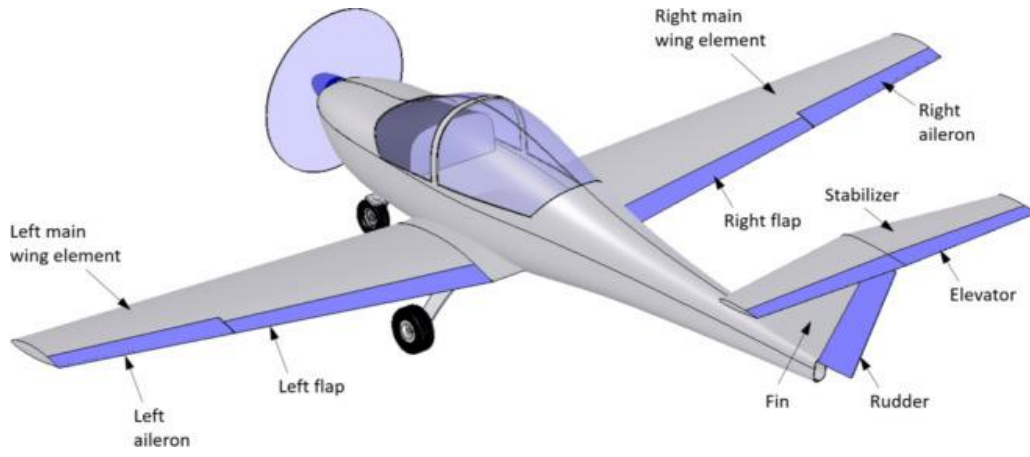


Figure 5: Aircraft control surfaces: ailerons, flaps, elevator, rudder, fin, and stabilizer (Source: Gudmundsson).

$$C_{M_{cg}} = C_{M_{ac,w}} + \left( C_{L_W} + C_{L_t} \frac{S_t}{S_w} \right) \frac{(\bar{x}_{cg} - \bar{x}_{ac,w})}{\bar{c}} - C_{L_t} V_H + C_{M_{ac,t}} V_H \frac{\bar{c}_t}{l_t} \quad (34)$$

Where the **horizontal tail volume coefficient** is defined as:

$$V_H = \frac{S_t \cdot l_t}{S_w \cdot \bar{c}} \quad (35)$$

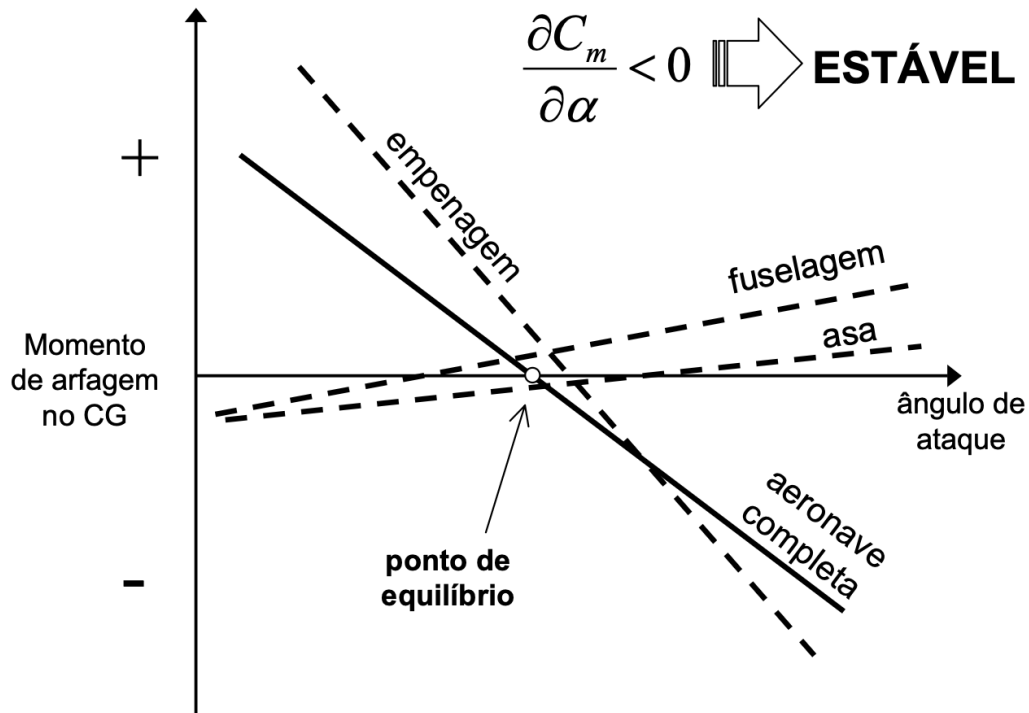


Figure 6: Pitching moment contributions from the tail (empenagem), fuselage, and wing. For static stability, the total aircraft curve must have a negative slope ( $\partial C_m / \partial \alpha < 0$ ).

### Neutral Point

The **neutral point** ( $x_n$ ) is the c.g. location at which the aircraft has **neutral static stability** ( $C_{M_\alpha} = 0$ ). It is defined by:

Table 14: Typical Static Margins for Various Aircraft

| Aircraft     | Static Margin |
|--------------|---------------|
| Cessna 172   | 0.19          |
| Learjet 35   | 0.13          |
| Boeing 747   | 0.27          |
| P-51 Mustang | 0.05          |
| F-16C        | 0.01          |
| X-29         | -0.33         |

$$\bar{x}_n = \bar{x}_{ac} + \frac{C_{L\alpha t}}{C_{L\alpha}} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) V_H \quad (36)$$

Where  $\partial \varepsilon / \partial \alpha$  is the downwash gradient at the tail.

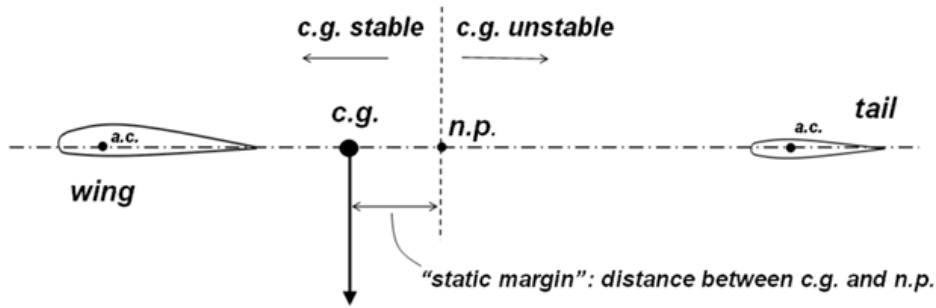


Figure 7: Static longitudinal stability: for a statically stable aircraft, the neutral point (n.p.) is aft of the center of gravity (c.g.). The static margin is the distance between them.

### Static Margin

The **static margin** (SM) is a measure of longitudinal stability:

$$SM = \bar{x}_n - \bar{x}_{cg} = -\frac{\partial C_M}{\partial C_L} \quad (37)$$

- **Larger SM**  $\rightarrow$  *more stable, less agile*
- **SM  $\approx 0$**   $\rightarrow$  *neutrally stable*
- **SM  $< 0$**   $\rightarrow$  *statically unstable*

### Scissor Plot ( $S_t/S$ )

The **scissor plot** is a design tool that plots the required tail-to-wing area ratio ( $S_t/S$ ) as a function of the c.g. location ( $\bar{x}_{cg}$ ). It establishes the forward and aft c.g. limits based on three criteria:

1. **Forward limit (stability):** ensures adequate static margin.
2. **Stall recovery:** ensures sufficient nose-down control authority at stall.
3. **Take-off rotation:** ensures sufficient nose-up moment to rotate during take-off.

The intersection of these constraints determines the minimum required tail size.



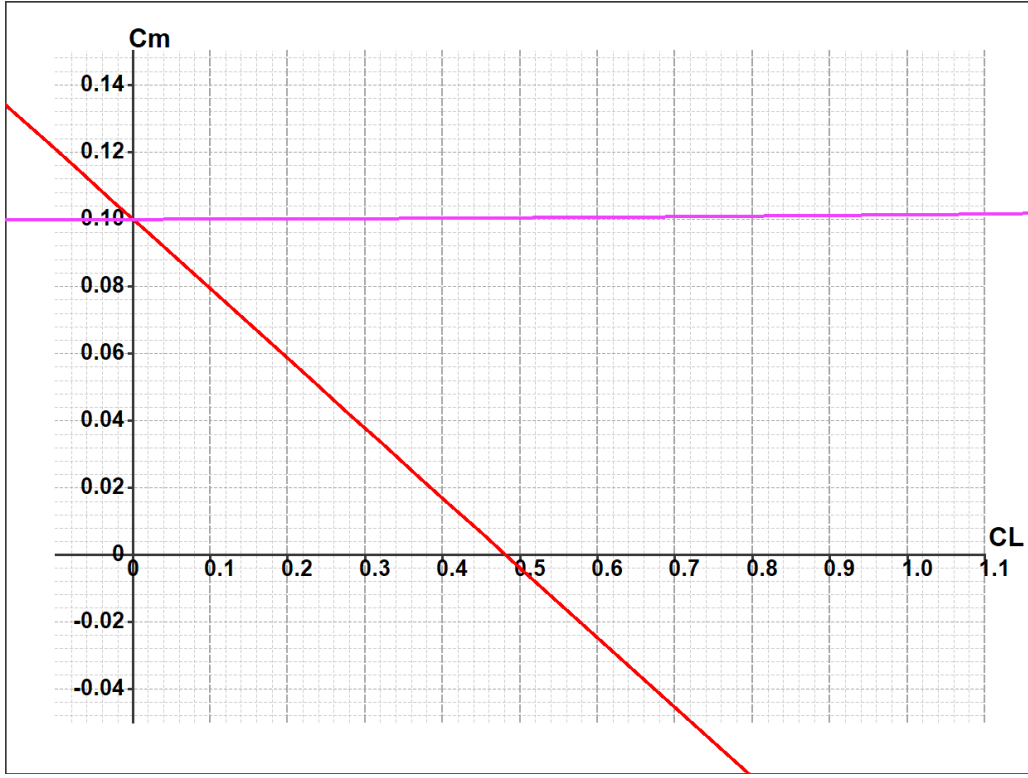


Figure 8: Example pitching moment ( $C_M$ ) vs. lift coefficient ( $C_L$ ) curve. The slope  $dC_M/dC_L$  is the negative of the static margin.

### Trim Condition

An aircraft is **trimmed** when it does not rotate about the c.g. in flight. Two conditions must be satisfied simultaneously:

$$C_M = 0 \quad \text{and} \quad C_L = C_{L_{trim}} = \frac{2W}{\rho V_\infty^2 S_w} \quad (38)$$

Two unknowns must be solved: the angle of attack  $\alpha_{trim}$  and the elevator deflection  $\delta_{e_{trim}}$ .

$$\alpha_{trim} = \frac{C_{M_0} C_{L_{\delta_e}} + C_{M_{\delta_e}} (C_{L_{trim}} - C_{L_0})}{C_{L_\alpha} C_{M_{\delta_e}} - C_{L_{\delta_e}} C_{M_\alpha}} \quad (39)$$

$$\delta_{e_{trim}} = -\frac{C_{M_0} C_{L_\alpha} + C_{M_\alpha} (C_{L_{trim}} - C_{L_0})}{C_{L_\alpha} C_{M_{\delta_e}} - C_{L_{\delta_e}} C_{M_\alpha}} \quad (40)$$

Where the elevator effectiveness derivatives are:

$$C_{L_{\delta_e}} = \frac{S_t}{S_w} C_{L_{\delta_{et}}}, \quad C_{M_{\delta_e}} = C_{L_{\delta_{et}}} \frac{S_t}{S_w} (\bar{x}_{cg} - \bar{x}_{ac,w}) - C_{L_{\delta_{et}}} V_H \quad (41)$$

### Trim Drag

Trim drag is the additional drag resulting from the deflection of the horizontal stabilizer and elevator to maintain the trim condition. Since the tail typically produces a downward force, the wing must produce extra lift to compensate, resulting in a slight increase in induced drag:

$$C_D = C_{D_0} + \frac{C_{L_w}^2}{\pi \cdot e_w \cdot AR_w} + \frac{C_{L_t}^2}{\pi \cdot e_t \cdot AR_t} \cdot \frac{S_t}{S_w} \quad (42)$$

**Suggested Reading**

- **Raymer:** Ch. 6
- **Gudmundsson:** Ch. 24 and 25
- **Drela:** Basic Aircraft Design Rules
- **Nelson:** Flight Stability and Automatic Control
- **Sadraey:** Ch. 5

